

Bayesian Trend Modeling and Forecasting of Argentina CPI

MS2506

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1 Introduction

This report analyzes the quarterly Consumer Price Index (CPI) time series for Argentina using Bayesian regression models. The CPI series exhibits extremely rapid growth towards the end (likely inflation), which motivates modeling on the log scale. We fit a Bayesian linear trend model as a baseline and a Bayesian spline trend model to capture the accelerating nonlinearity observed in the late 1980s. The model fit is evaluated using convergence diagnostics, posterior predictive checks (PPC), and residual autocorrelation checks. Finally, we generate a five quarter near future forecast using the superior spline model to capture the quickly accelerating trajectory.

2 Data

We have chosen the **ArgentinaCPI** from the **AER** package, which is a quarterly univariate time series from 1970Q1 to 1989Q4, indexed so that 1969Q4 = 1. The dataset contains 80 observations (with frequency = 4). The raw CPI is highly right-skewed and grows by orders of magnitude, so we transform to $\log(\text{CPI})$ before modeling.

2.1 Exploratory plots

Figures 1 and 2 show the raw CPI series and its distribution, along with figures 3 and 4 for the log-transformed series and distribution. The log scale produces a more interpretable, somewhat trend like structure, while still showing changes in slope over time.

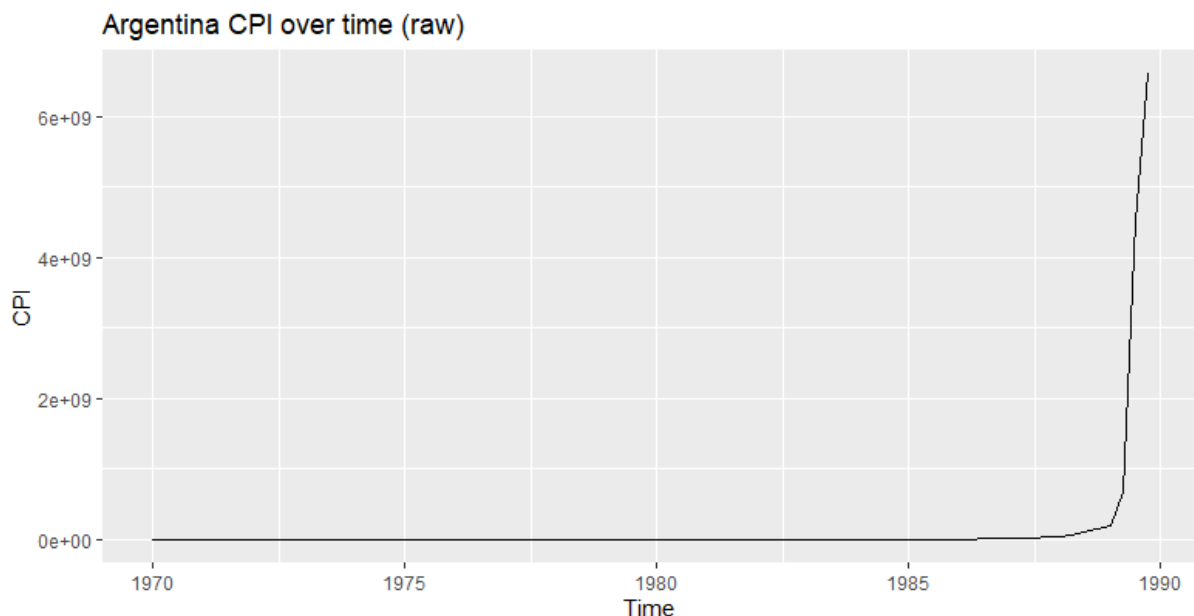


Figure 1: Argentina CPI over time (raw scale).

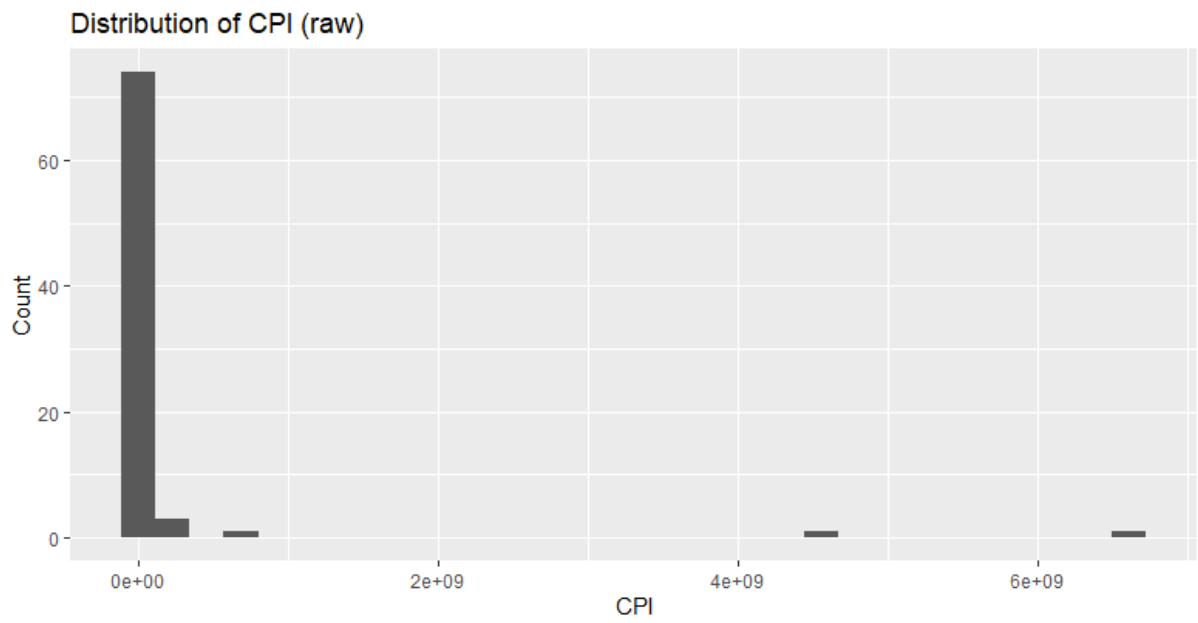


Figure 2: Distribution of CPI (raw scale).

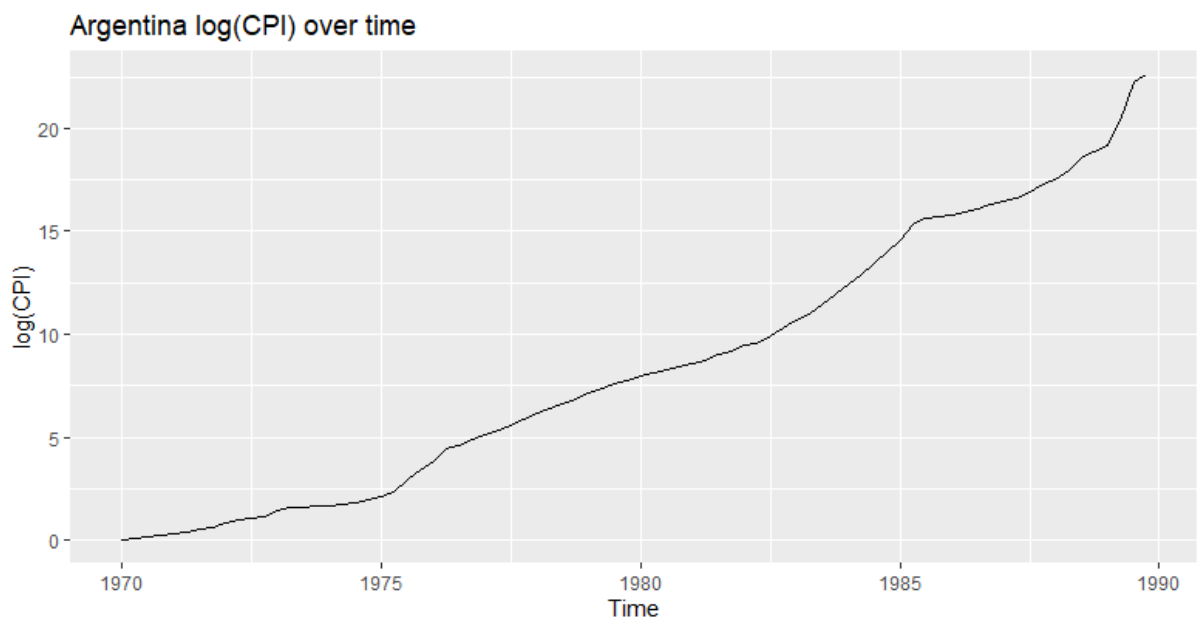


Figure 3: Argentina log(CPI) over time.

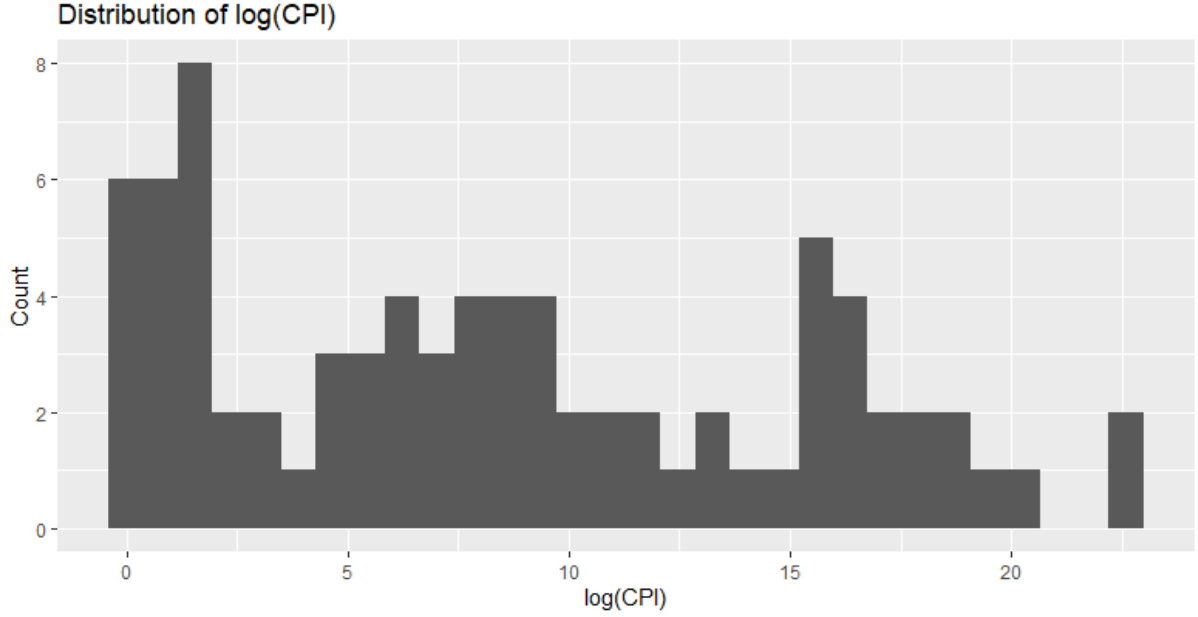


Figure 4: Distribution of $\log(\text{CPI})$.

3 Method

3.1 Data transformation and preprocessing

The raw CPI data shows exponential growth and skews to the right, making direct modeling numerically unstable. To fix this, we first transformed the dependent variable to the natural logarithmic scale ($\log(\text{CPI})$). We scaled the data to have a mean of zero and a standard deviation of one. This standardization allows for the use of standard weakly informative priors without requiring manual tuning based on the scale of the billions of pesos observed. All final forecasts and slope estimates were back transformed to the original CPI scale for interpretation.

3.2 Model 1: Bayesian linear trend

Our baseline model is a Bayesian linear regression fitted using the `brms` package in R. We assumed a Gaussian likelihood for the standardized data. To constrain the posterior we assigned weakly informative Normal priors to the intercept and slope coefficients, and a Student-t prior to the residual standard deviation. To ensure the reliability of our results, we performed a prior sensitivity analysis. We re-fitted the linear model using significantly wider priors (doubling the variance) to verify that the estimated growth rates were robust and not artificially affected by our initial prior choice.

3.3 Model 2: Bayesian spline trend

The linear model assumes a constant growth rate, which contradicts the visual evidence of accelerating hyperinflation in the late 1980s. To capture this nonlinearity, we extended the analysis using a smoothing spline. This was implemented using the `s()` function from `brms`, which allows the trend to change smoothly over time rather than following a straight

line. The initial fit of the spline model resulted in a small number of divergent transitions, a diagnostic warning indicating potential bias in the posterior approximation. To resolve this, we adjusted the sampler’s control parameters in the code by increasing `adapt_delta` to 0.995 and the `max_treedepth` to 15. These adjustments forced the sampler to take smaller steps, eliminating the divergences and ensuring valid inference.

4 Computation and diagnostics

All models are fitted in `brms` with 4 chains. For the models we used 4000 iterations with 1000 warmup.

4.1 Linear model convergence and PPC

The linear model shows good sampling diagnostics (all $\hat{R} \approx 1$, large effective sample sizes). Figure 5 shows trace/density summaries. Posterior predictive checks are shown in Figures 6–7 and suggest the model captures the overall distribution of standardized $\log(\text{CPI})$ reasonably well.

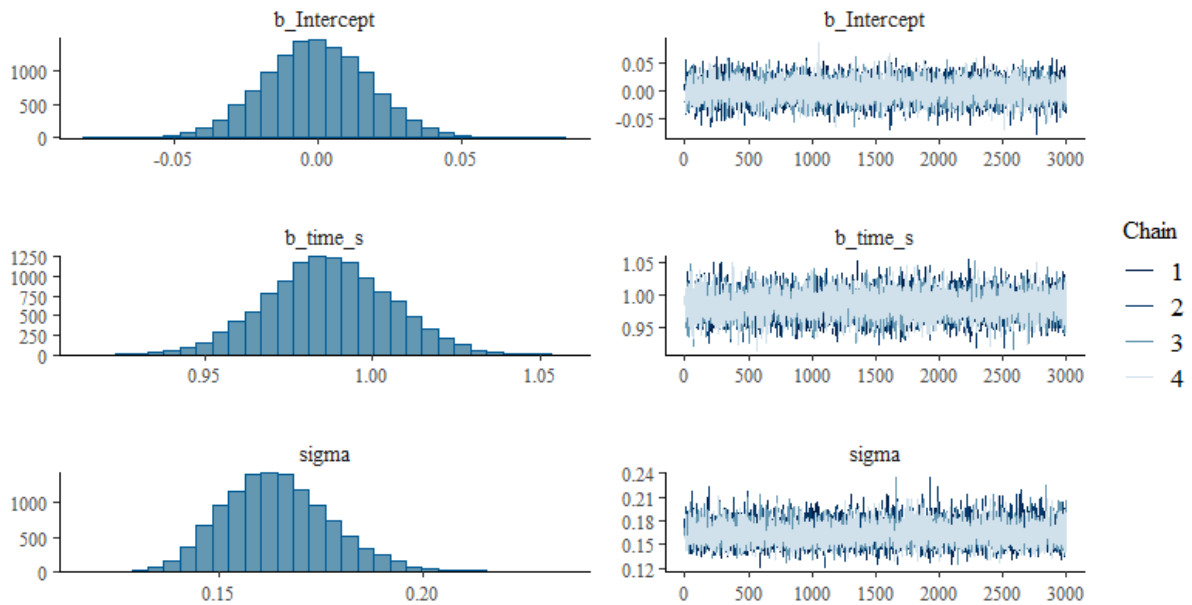


Figure 5: Linear model parameter summaries (posterior histograms and trace plots).

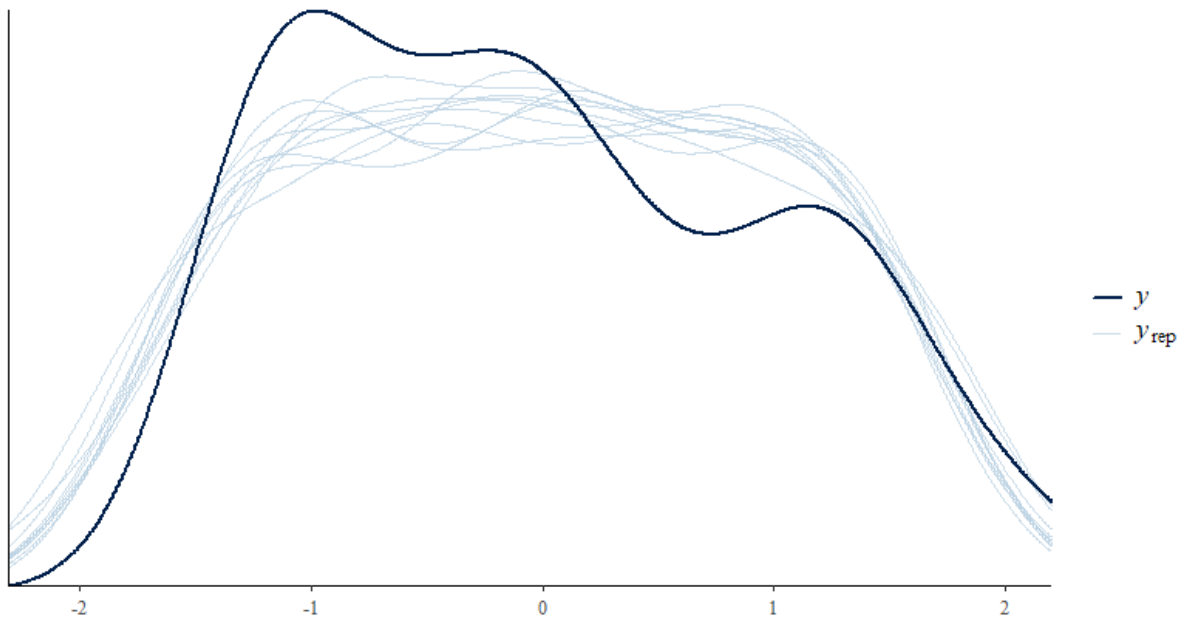


Figure 6: Linear model PPC: density overlay (standardized scale).

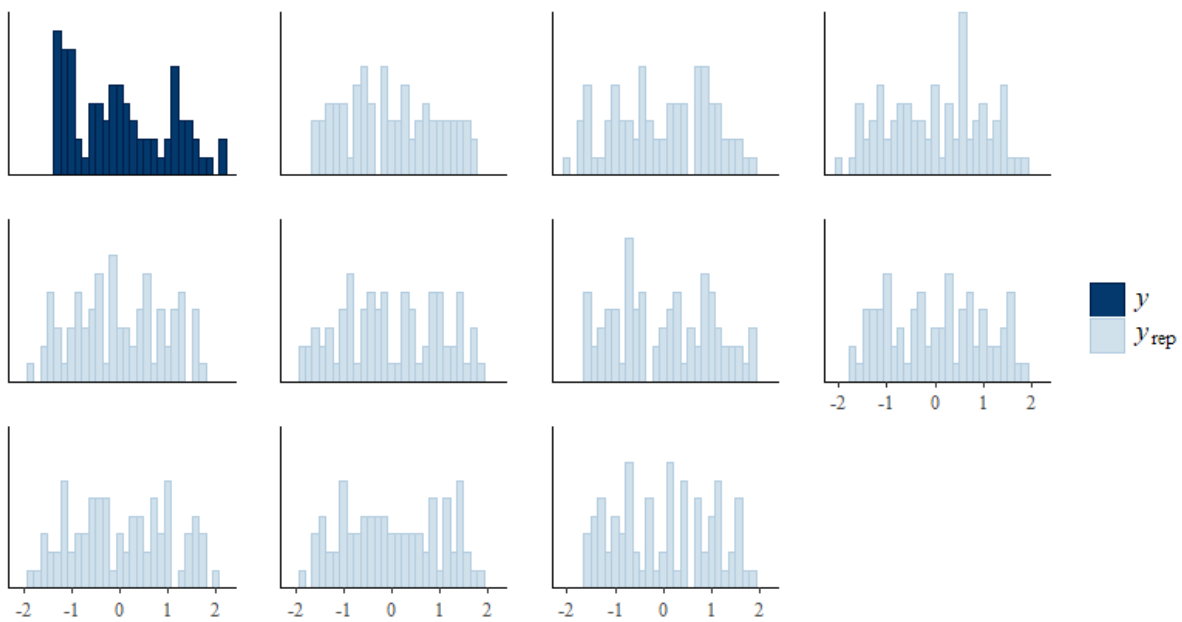


Figure 7: Linear model PPC: histogram checks (standardized scale).

4.2 Fitted mean, residuals, and autocorrelation

Figure 8 compares the observed $\log(\text{CPI})$ and the fitted mean from the linear trend model (back on the original $\log$ scale). The linear model follows the overall trend but cannot track changes in slope perfectly (for example the 1980s acceleration). A histogram of residuals is shown in Figure 9. We also check residual autocorrelation (Figure 10). The

ACF shows clear positive correlation at small lags, indicating remaining serial dependence not captured by an iid error model. It also reaches zero, albeit a little slow.

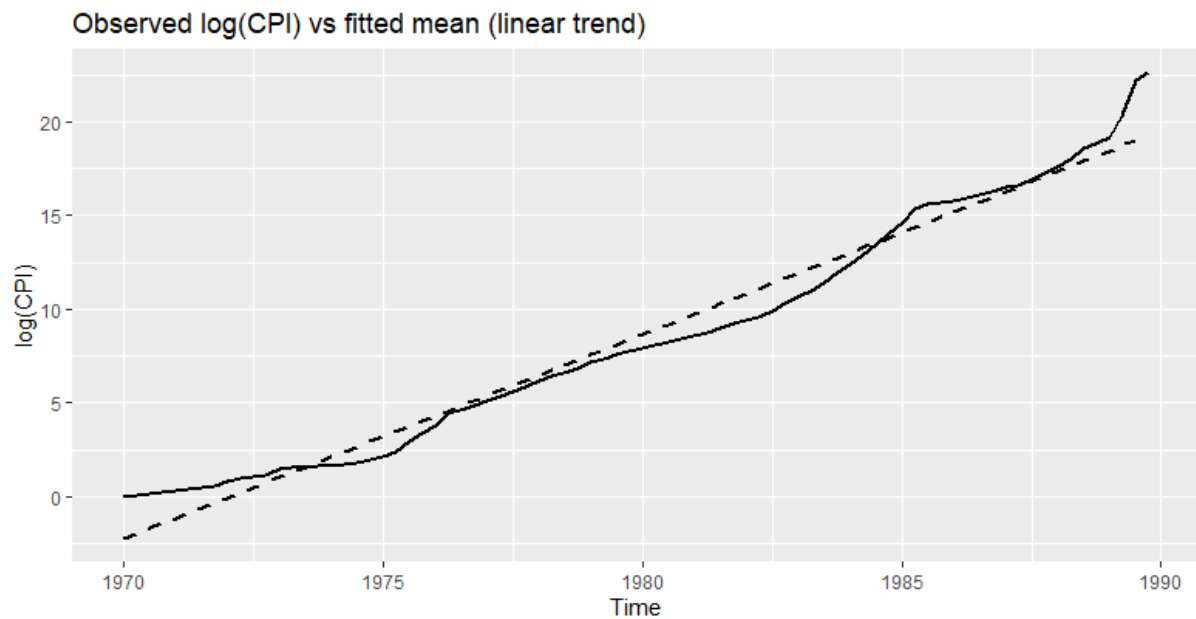


Figure 8: Observed $\log(\text{CPI})$ vs fitted mean (linear trend).

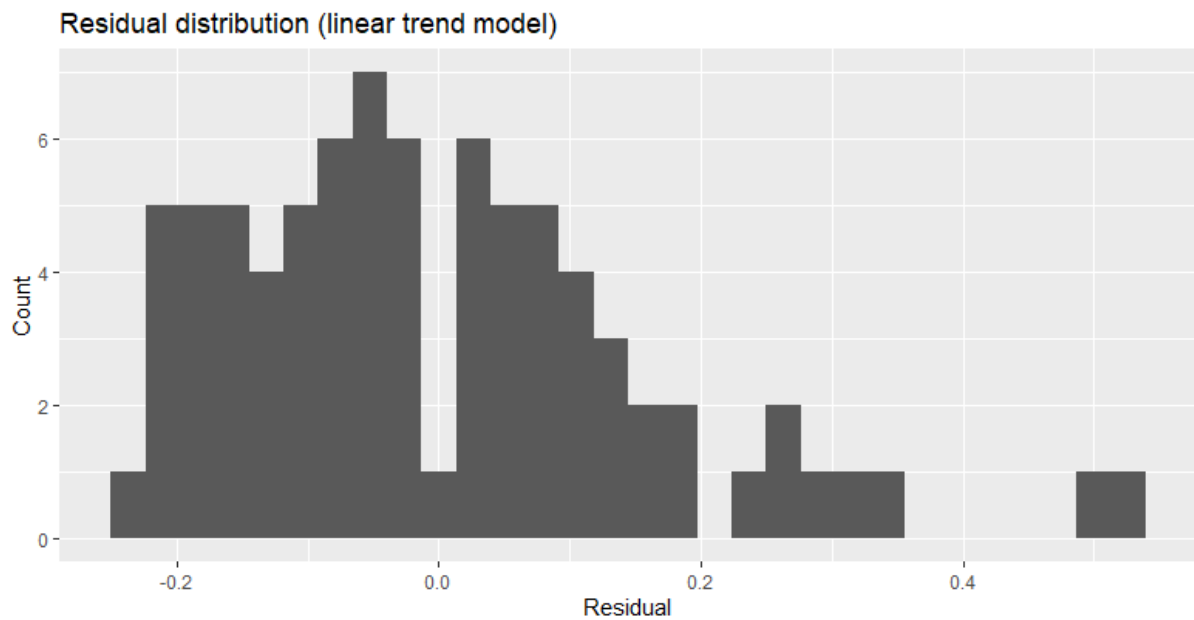


Figure 9: Residual distribution (linear trend model; standardized residuals).

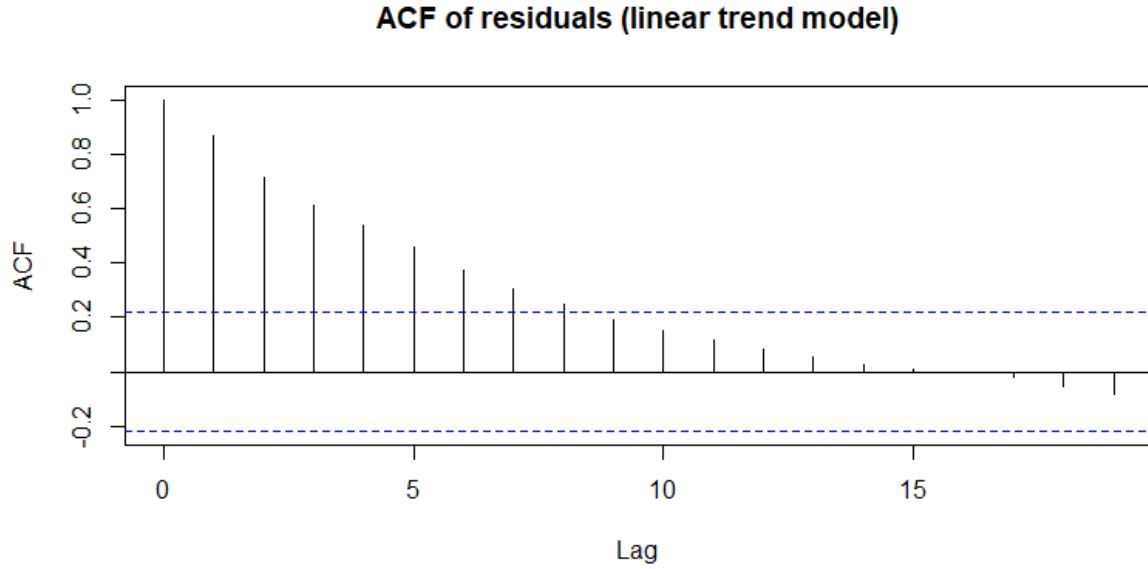


Figure 10: ACF of residuals (linear trend model). Spikes at small lags suggest remaining serial correlation.

4.3 Spline model diagnostics

After raising `adapt_delta` and `max_treedepth`, the spline model had zero divergent transitions and good convergence diagnostics. Posterior predictive checks are shown in Figures 11–12. The spline can adapt to changes in slope and typically fits the mean structure more closely than a straight line.

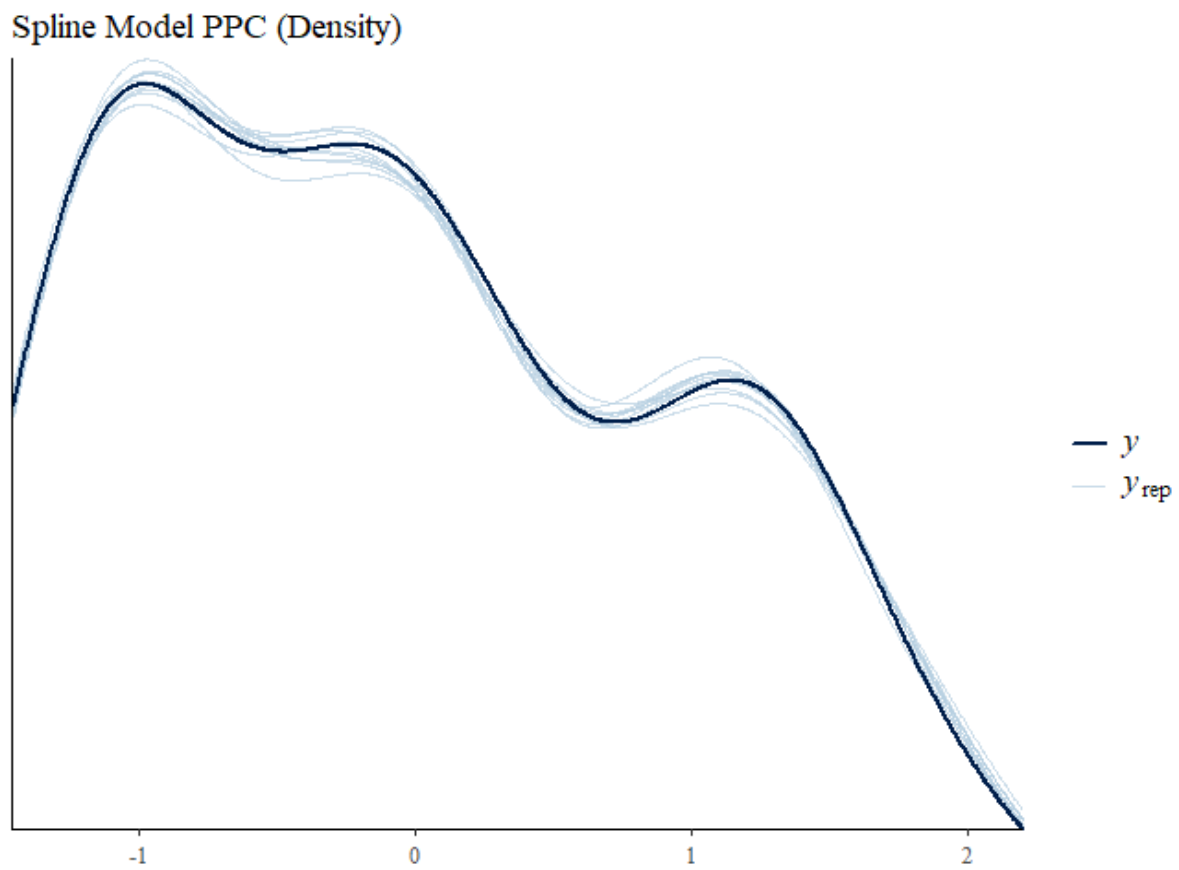


Figure 11: Spline model PPC: density overlay (standardized scale)

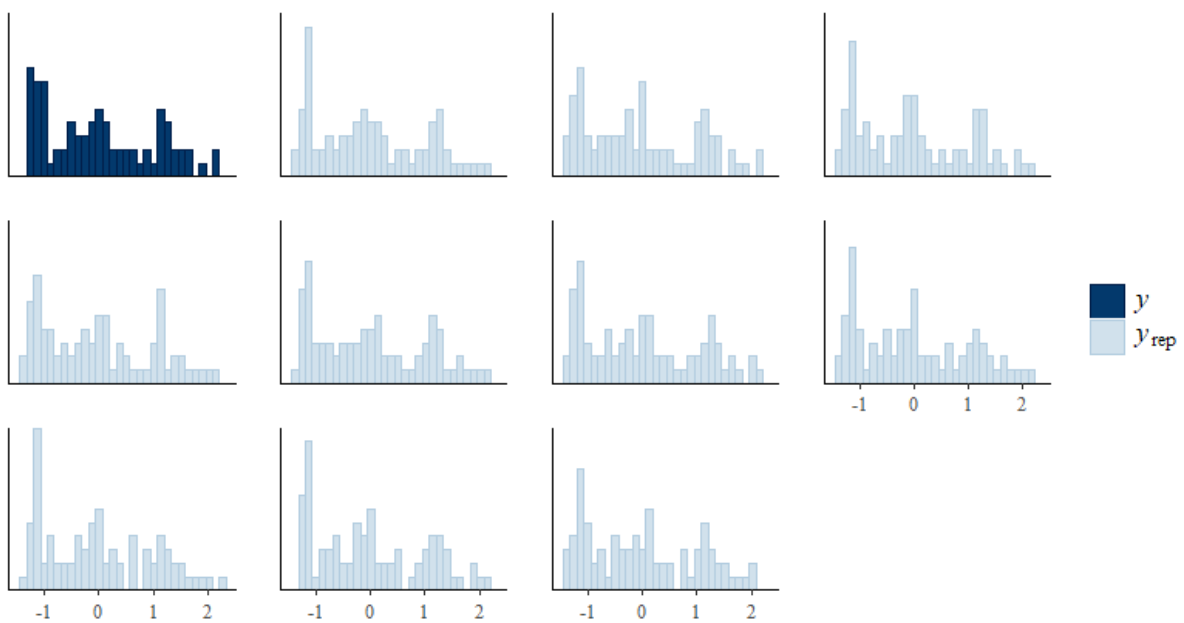


Figure 12: Spline model PPC: histogram checks (standardized scale).

5 Results

5.1 Linear trend parameter estimates

Table 1 summarizes the key posterior estimates from the linear model (standardized scale). The slope is strongly positive.

Parameter	Mean	SD	95% CI (low)	95% CI (high)
Intercept	-0.00	0.02	-0.04	0.04
β (time.s)	0.99	0.02	0.95	1.02
σ	0.16	0.01	0.14	0.19

Table 1: Linear model posterior summary (standardized scale) from `brms`.

5.2 Interpretation on the original log scale

Back transforming the slope to the original $\log(\text{CPI})$ scale gives an estimated slope per year with posterior median 1.0879 and 95% interval [1.0477, 1.1283]. Interpreted as multiplicative growth, this corresponds to approximately 196.8% CPI growth per year (95% interval roughly 185.1% to 209.0%), or about 31.25% growth per quarter (95% interval roughly 29.94% to 32.59%). The posterior probability that the slope is positive is essentially 1.

5.3 Sensitivity to priors

Using wider priors ($\alpha, \beta \sim \mathcal{N}(0, 2)$, $\sigma \sim t_3(0, 2)$) produced nearly identical growth estimates where median yearly growth is about 196.9% with a very similar 95% interval. This indicates the likelihood is highly informative for the trend parameter under these reasonable weakly-informative prior choices.

5.4 Linear vs spline fitted mean

Figure 13 compares fitted mean curves on the $\log(\text{CPI})$ scale. The spline better captures changes in slope (nonlinear trend), especially around periods where inflation accelerates. However, the linear model remains a useful baseline for summarizing an average exponential growth rate over the full period.

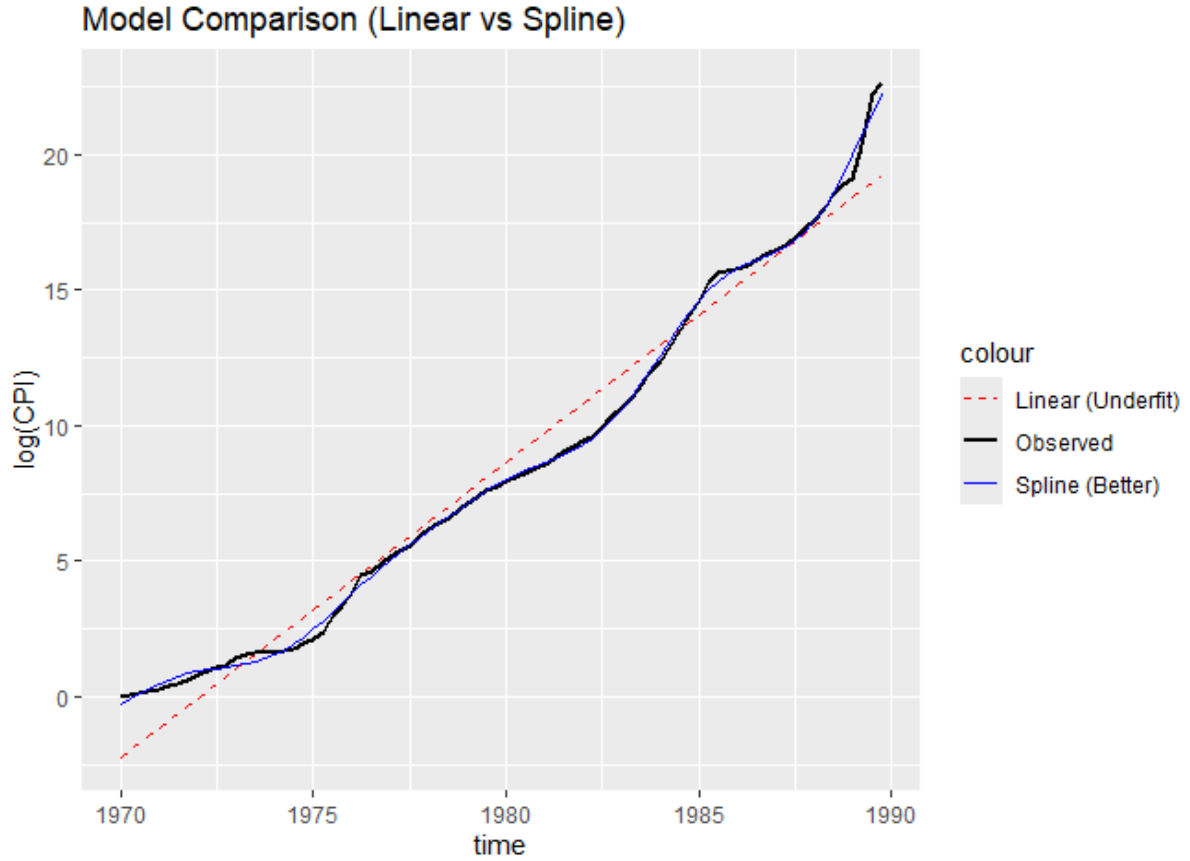


Figure 13: Observed $\log(\text{CPI})$ vs fitted means (linear vs spline).

6 Forecasting

We forecast the next $h = 5$ quarters beyond 1989Q4 using the **spline trend model**. The linear model was deemed unsuitable for forecasting due to its inability to capture the quickly accelerating inflation trend observed at the end of the series. Two types of intervals are reported. Predictive intervals (from `posterior_predict`) represent uncertainty in a future observation and include observation noise, resulting in wider bounds. Mean intervals (from `posterior_epred`) show uncertainty in the underlying conditional mean and are narrower. Table 2 shows the forecasts on the CPI scale. Note that due to the hyperinflation captured by the spline, the forecasted values grow rapidly into the billions. Figure 14 visualizes the observed CPI and forecasts, where the dotted line is the mean forecast, and colored area is the 95% bounds.

Time	Forecast Mean	Lower 95% (Predictive)	Upper 95% (Predictive)
1990 Q1	9,978,795,444	5,481,854,056	17,414,005,923
1990 Q2	21,819,319,004	11,562,917,140	39,259,807,964
1990 Q3	47,762,035,356	24,068,382,332	88,504,390,505
1990 Q4	104,665,457,470	50,617,949,572	203,819,993,720
1991 Q1	229,616,565,080	102,947,110,313	462,728,203,920

Table 2: Forecasts from the Bayesian Spline Model (Predictive Intervals).

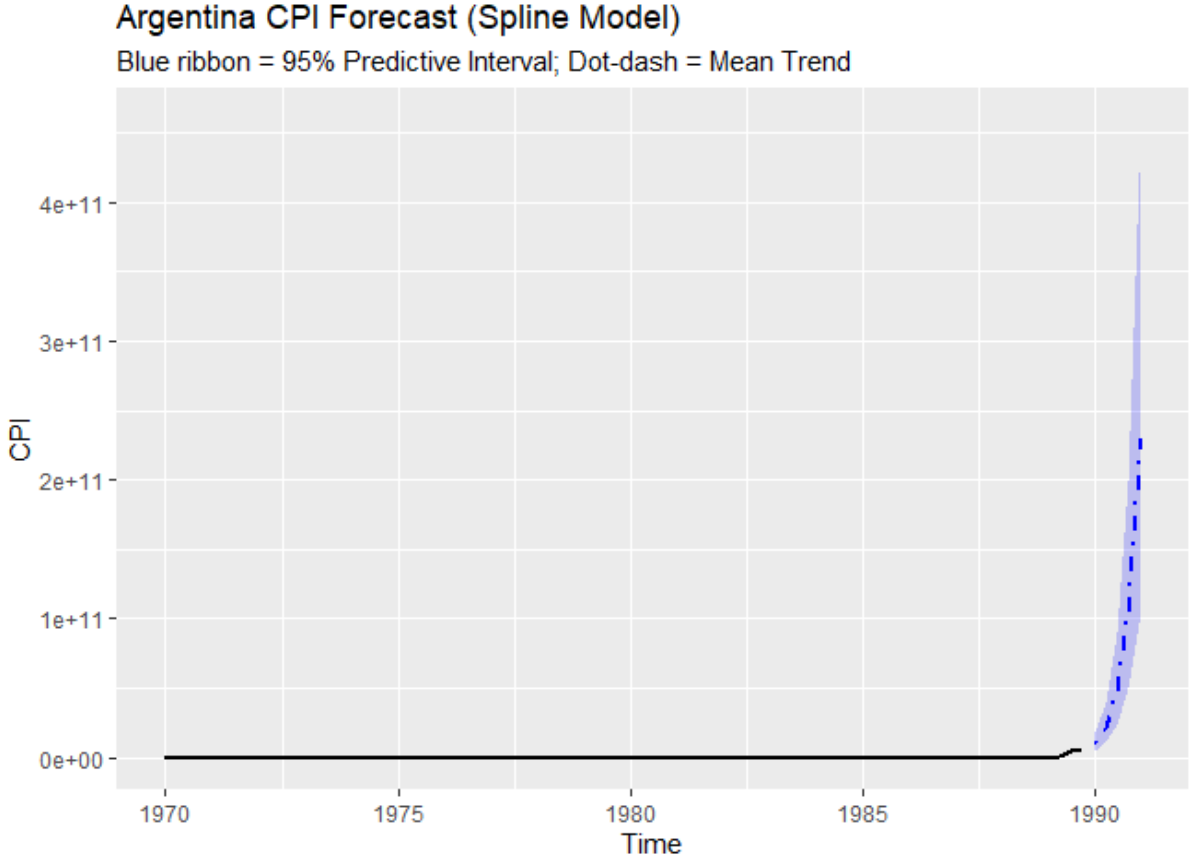


Figure 14: Spline model forecast plot for 1990Q1 to 1991Q1

7 Discussion and limitations

The log transformation is important since raw CPI is quite skewed and grows fast, while $\log(\text{CPI})$ yields a roughly trend-like structure suitable for regression. The linear trend model provides a clear average growth estimate but fails to capture the accelerating inflation in the late 1980s. Diagnostics indicate limitations typical for time series. Residual autocorrelation remains substantial (ACF spikes at small lags), violating the iid error assumption. The spline model fits the mean structure much more flexibly, capturing the inflation spike that the linear model misses. Thus, the spline model provides far more realistic forecasts for 1990, projecting CPI values in the billions, whereas a linear

model would have erroneously predicted a reversion to the long-term average. This is also expected compared to real life where Argentina had inflation rates up to about 3000% around 1990 and if economic changes hadn't taken place the inflation acceleration would likely have continued in a similar way as the forecast [1]

8 Conclusion

A Bayesian linear trend model on standardized $\log(\text{CPI})$ estimates rapid average growth in Argentina CPI over 1970–1989. However, the linear model fails to track the hyperinflation acceleration at the end of the 80s. A spline trend model significantly improves the fit by capturing this nonlinearity. Forecasts generated from the spline model project continued explosive growth, with mean estimates reaching approximately 229 billion by early 1991. This shows the lacking aspects of linear models in nonlinear scenarios.

References

- [1] The World Bank (2025). *Inflation, GDP deflator (annual %) - Argentina*. World Bank Open Data. <https://data.worldbank.org/indicator/NY.GDP.DEFL.KD.ZG?locations=AR>